Chromatic Admissible Contexts and Enforced Completeness of Nonadjacent Vertices

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In a d-uniform quantum hypergraph, every measurement context is rainbow in every strong d-coloring. We study *chromatic admissible contexts*: sets of d pairwise nonadjacent vertices that receive every color exactly once in all such colorings. Equivalently, their state sum is one on all extendable two-valued states. The stronger notion of a *two-valued state admissibility context* requires the same normalization on every two-valued state. We construct a faithful 53-ray, 34-context example in $\mathbb{R}^3$ by pasting two nonseparating Kochen–Specker gadgets at one ray. Its two independent admissible triples are normalized on all 224 two-valued states and are rainbow in all 768 labeled strong 3-colorings. A short implication proof establishes this universal normalization, whereas an explicit Born preparation gives total 8/9 on one triple. Admissibility is therefore distinct from affine and operator normalization. It also differs from Navara–Svozil pseudocontexts, whose balanced affine certificates equate two probability sums and imply color-multiset equivalence. Within the examined Yu–Oh restrictions, independent chromatic admissible triples first occur at $H_{20,12}$, with two triples; $H_{21,13}$ has 21, classified through its two color signatures. Structural color propagation proves universal completeness of its hub, core, and coplanar triples. Cascaded roof-twin gadgets $G_{41,25}$ and $G_{41,26}$ give further two-valued admissibility contexts, with a two-step binary proof and separately verified faithful coordinates. Explicit binary filters reduce the state spaces of $H_{20,12}$ and $H_{21,13}$ to their extendable states. Further real and complex examples exhibit universal rainbow subsets, chromatic twins, and color-multiset-equivalent pairs. A census of 19 faithfully represented configurations separates these assignment-level relations from affine conservation laws and identifies the state family on which admissibility holds.

I. INTRODUCTION

A complete rank-one projective measurement in $\mathbb{F}^d$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$) is represented by an orthonormal basis of rays $\{[q_v] : v \in e\}$ whose rank-one projectors $P_v = q_v q_v^\dagger / \|q_v\|^2$ resolve the identity,

$$\sum_{v \in e} P_v = I_d. \tag{1}$$

For any finite d-uniform hypergraph $H = (V, E)$ of intertwined projective measurements ($n = |E|$), every admissible probability assignment $p : V \to [0, 1]$—whether generated by the quantum Born rule $p(v) = \mathrm{Tr}(\rho P_v)$, by convex combinations of classical two-valued states $s : V \to \{0, 1\}$ in $\mathcal{S}(H)$, or by more general normalized weight assignments—satisfies the context normalization equations

$$\sum_{v \in e} p(v) = 1 \qquad (\forall\, e \in E). \tag{2}$$

We write $\mathcal{W}(H)$ for all normalized weights $p : V \to [0, 1]$ satisfying Eq. (2), and $\mathcal{S}(H) = \mathcal{W}(H) \cap \{0, 1\}^V$ for its two-valued states. The entire real affine solution space is

$$\mathcal{A}(H) = \Big\{x \in \mathbb{R}^V : \sum_{v \in e} x(v) = 1 \text{ for all } e \in E\Big\}. \tag{3}$$

These three domains must be distinguished: an identity on $\mathcal{S}(H)$ need not be an identity on $\mathcal{W}(H)$ or on $\mathcal{A}(H)$.

In Refs. [1, 2], Navara and Svozil introduced *pseudocontexts* (or *pseudoblocks*). Under the restrictive affine definition of the latter work, these are disjoint sets $A, B \subseteq V$ of equal size $k \geq 2$, each independent in the two-section graph $\Gamma(H)$, whose probability sums are forced to coincide by a balanced linear combination of context equations. In particular,

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b) \qquad (\forall\, p \in \mathcal{A}(H)). \tag{4}$$

The common sum need not equal one, and neither set is required to be complete or exclusive under a coloring. When H admits a faithful orthogonal representation (FOR) [3, 4] in $\mathbb{F}^d$, the affine certificate forces the operator identity $\sum_{a \in A} P_a = \sum_{b \in B} P_b$ in every FOR of H. That operator identity imposes geometric rigidity [2, 5]:

1. *No single complete nonorthogonal d-tuple:* If $U = \{u_1, \ldots, u_d\}$ satisfies $\sum_{u \in U} P_u = I_d$ in $\mathbb{C}^d$, then $P_{u_i} P_{u_j} = 0$ for $i \neq j$, so U must be an orthogonal basis rather than a nonorthogonal set.

2. *No size-one ($k = 1$) Hilbert-space pseudocontext:* $P_a = P_b$ implies $[q_a] = [q_b]$, violating ray injectivity.

3. *No size-two ($k = 2$) real Hilbert-space pseudocontext:* In $\mathbb{R}^d$, if $q_{a_1} \not\perp q_{a_2}$, the spectral decomposition of $P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}$ uniquely fixes the unordered pair of rays $\{[q_{a_1}], [q_{a_2}]\} = \{[q_{b_1}], [q_{b_2}]\}$. Thus paired affine pseudocontexts require $k \geq 2$ over $\mathbb{C}$ (attained by a 20-ray octagon H_{20} in $\mathbb{C}^3$) and $k \geq 3$ over $\mathbb{R}$ (attained by Mayet's 15-ray hypergraph $M_{15,8}$ in $\mathbb{R}^3$) [2].

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A parallel line of research studies *common-spectrum spectral assignments* (strong d-colorings) of quantum hypergraphs [6–9]. Assigning each ray $v \in V$ a context-independent eigenvalue $\lambda_{c(v)}$ from a fixed d-element spectrum $\Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$ defines nondegenerate commuting observables $A_e = \sum_{v \in e} \lambda_{c(v)} P_v$ [10, 11] if and only if $c : V \to \{0, \ldots, d-1\}$ is a strong d-coloring ($c \in \mathcal{C}_d(H)$), injective on every context $e \in E$. In the companion work [9], strong d-colorings were characterized by $n \times d$ *Context–Color Matrices* $M \in V^{n \times d}$ whose rows permute the d rays of each context and whose d columns are the *extendable* two-valued states $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$.

The central question of this paper is different: can a set of pairwise nonadjacent vertices be forced to be complete and exclusive by all global strong d-colorings? We call such a set a *chromatic admissible context*. Its vertices occupy all d columns of every Context–Color Matrix, exactly once each. More generally, an *admissibility context relative to a nonempty state family* is an independent set whose state sum is identically one on that family. This formulation separates the chromatic case, based on $\mathcal{S}_{\text{ext}}(H)$, from the stronger two-valued case, based on all of $\mathcal{S}(H)$. Neither formulation presupposes an affine or operator identity. Alongside this central notion, we study universal rainbow exclusivity, chromatic twins, and paired color-multiset equivalence. We show that:

- A faithful real three-dimensional construction $H_{53,34}$ has two independent triples normalized on all 224 two-valued states. Pasting two nonseparating gadgets gives a short structural proof; an explicit Born preparation with total 8/9 separates two-valued state admissibility from affine and operator normalization (Theorem 6).
- Every balanced affine pseudocontext (A, B) automatically enforces *color-multiset conservation* $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ in every $n \times d$ Context–Color Matrix (Theorem 3 and Corollary 4).
- Faithfully coordinatized real three-dimensional hypergraphs admit both *chromatic admissible contexts* (normalized on $\mathcal{S}_{\text{ext}}(H)$, starting at $H_{20,12}$ and $H_{21,13}$) and genuine *two-valued state admissibility contexts* (normalized on all of $\mathcal{S}(H)$, realized in $G_{41,25}$ and $G_{41,26}$ via cascaded roof-twin gadgets). The examples also exhibit non-twin color-multiset-equivalent pairs of size two, universally rainbow nonorthogonal subsets, and chromatic twin classes, with related phenomena in $\mathbb{R}^4$. Because $\{0,1\}$-integrality on Specker bugs enforces biconditional two-valued twins $s(x) = s(x')$ that do not hold on $\mathcal{A}(H)$ ($\text{aff}\,\mathcal{S}(H) \subsetneq \mathcal{A}(H)$), the operator rigidity statements above remain valid.

Section II defines admissibility relative to a state family and the related chromatic relations. Section III analyzes affine pseudocontexts and their color-multiset consequences. Section IV constructs a two-valued state admissibility context in $\mathbb{R}^3$, with its structural proof, full hypergraph drawing, and exact FOR. Section V then treats chromatic admissible contexts and color-multiset-equivalent pairs, including structural color propagation in the Yu–Oh example. Section V D develops the roof-twin cascades $G_{41,25}, G_{41,26}$ and binary state restrictions in the Yu–Oh examples. Section VI treats four-dimensional relations, and Section VII gives the census and conclusion.

II. CONTEXT–COLOR MATRICES AND ADMISSIBILITY CONTEXTS

Let $H = (V, E)$ be a finite d-uniform hypergraph with $V = \bigcup_{e \in E} e$ and $n = |E|$ contexts $E = \{C_1, \ldots, C_n\}$. Its two-section graph $\Gamma(H)$ has vertex set V and edges $\{u, v\}$ whenever $u \neq v$ share at least one context $C_i \in E$. A *faithful orthogonal representation* (FOR) in $\mathbb{F}^d$ assigns nonzero vectors $q_v \in \mathbb{F}^d$ to $v \in V$ such that $[q_u] \neq [q_v]$ for $u \neq v$ and $\langle q_u, q_v \rangle = 0$ if and only if $\{u, v\} \in E(\Gamma(H))$. Thus an independent set $U \subseteq V$ in $\Gamma(H)$ is a family of *pairwise nonorthogonal* rays.

By the Universal $n \times d$ Context–Color Matrix Criterion [9, Theorem 1], H admits a strong d-coloring $c \in \mathcal{C}_d(H)$ if and only if the vertices of each context C_i can be ordered into an $n \times d$ matrix $M = (M_{i,j}) \in V^{n \times d}$ ($1 \leq i \leq n$, $0 \leq j \leq d-1$) such that (1) row i is a permutation of C_i, and (2) each vertex $v \in V$ appears in a single column $j = c(v)$. The column vertex sets $S_j = \{M_{i,j} : 1 \leq i \leq n\} = c^{-1}(\{j\})$ form d pairwise disjoint two-valued state supports whose indicator functions $s_j = \mathbf{1}_{S_j} \in \mathcal{S}_{\text{ext}}(H)$ partition V:

$$\sum_{j=0}^{d-1} s_j(v) = 1 \qquad (\forall\, v \in V). \tag{5}$$

Throughout, N_c counts normalized strong colorings obtained by fixing the color order on one reference context. Every labeled strong d-coloring has a unique such normalization and is recovered by one of the $d!$ color permutations. All universal color relations below are invariant under those permutations, so normalized representatives cover every strong d-coloring of the prescribed palette.

In every Context–Color Matrix, each actual context is a complete rainbow row. The following definition isolates the same exclusivity and completeness when they are enforced on vertices that share no context with one another.

Definition 1 (Admissibility Context Relative to a State Family)**.** Let $\varnothing \neq \mathcal{T} \subseteq \mathcal{S}(H)$. An independent set $U \subseteq V$ in $\Gamma(H)$ is an *admissibility context relative to* $\mathcal{T}$ if

$$\sum_{u \in U} s(u) = 1 \qquad (\forall\, s \in \mathcal{T}). \tag{6}$$

The equation enforces both exclusivity (at most one vertex is assigned one) and completeness (at least one is assigned

one), despite pairwise nonadjacency. For $\mathcal{T} = \mathcal{S}_{\text{ext}}(H) \neq \varnothing$, we call U a *chromatic admissible context*; for $\mathcal{T} = \mathcal{S}(H) \neq \varnothing$, we call it a *two-valued state admissibility context*. The state family is part of the definition. No affine certificate or projector normalization is assumed.

When $\mathcal{C}_d(H) \neq \varnothing$, every two-valued state admissibility context is a chromatic admissible context because $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$. The converse need not hold. The chromatic case has exactly d vertices: for any coloring, sum Eq. (6) over its d color-class states and use Eq. (5) to obtain $|U| = d$. Theorem 3(1) identifies this condition with universal rainbow completeness.

We now distinguish this normalization property from the other universal chromatic relations used below.

Definition 2 (Admissibility and Related Chromatic Relations)**.** Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$.

1. *Class I: Chromatic Admissible Contexts.* An independent set $U = \{u_1, \ldots, u_d\} \subseteq V$ is a chromatic admissible context precisely when it receives all d colors in every strong d-coloring:

$$\big\{c(u_1), \ldots, c(u_d)\big\} = \{0, 1, \ldots, d-1\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{7}$$

   If a non-context d-tuple satisfies Eq. (7) but has both adjacent and nonadjacent pairs, we call it a *partially nonorthogonal complete rainbow d-tuple*. It is not an admissibility context under Definition 1, which requires pairwise nonadjacency.

2. *Class II: Universal Rainbow Sets and Chromatic Twin Classes.* Let $U = \{u_1, \ldots, u_k\} \subseteq V$ be an independent set with $k \geq 2$.

   (a) U is a *universally rainbow (chromatically exclusive) k-tuple* ($2 \leq k < d$) if its elements receive pairwise distinct colors in every strong d-coloring:

$$c(u_a) \neq c(u_b) \quad (1 \leq a < b \leq k), \qquad \forall\, c \in \mathcal{C}_d(H). \tag{8}$$

   (b) Write $u \sim_c v$ if $c(u) = c(v)$ for every $c \in \mathcal{C}_d(H)$; such vertices are *chromatic twins*. A *chromatic twin class* is an equivalence class of $\sim_c$, and a *universally monochromatic set* is any subset of such a class. Thus U is universally monochromatic (with no upper bound on k) if

$$c(u_1) = c(u_2) = \cdots = c(u_k) \qquad \forall\, c \in \mathcal{C}_d(H). \tag{9}$$

3. *Class III: Color-Multiset-Equivalent Pairs.* Two disjoint independent sets $A = \{a_1, \ldots, a_k\}$ and $B = \{b_1, \ldots, b_k\}$, with $2 \leq k \leq d$, form a *color-multiset-equivalent pair* if

$$\big\{\{c(a) : a \in A\}\big\} = \big\{\{c(b) : b \in B\}\big\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{10}$$

   The pair is *non-twin* if no element of A is a chromatic twin of an element of B ($\neg(a_i \sim_c b_j)$ for all $a_i \in A$, $b_j \in B$). Color-multiset equivalence alone does not impose exclusivity or completeness on either side. We reserve *affine pseudocontext* for a pair with a balanced context certificate. In the census, we count non-twin pairs and omit pairs in which both sides are already chromatic admissible contexts; their equality follows directly from the single-set normalizations.

Theorem 3 (Chromatic Relations via Extendable State Sums)**.** *Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$ and extendable state set $\mathcal{S}_{\text{ext}}(H) = \{s_j = \mathbf{1}_{c^{-1}(\{j\})} : c \in \mathcal{C}_d(H),\ 0 \leq j \leq d-1\}$. Then:*

1. *A d-element set $U = \{u_1, \ldots, u_d\} \subseteq V$ satisfies $\{c(u) : u \in U\} = \{0, \ldots, d-1\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u \in U} s(u) = 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{11}$$

2. *A k-element set $U = \{u_1, \ldots, u_k\} \subseteq V$ ($2 \leq k < d$) satisfies $c(u_a) \neq c(u_b)$ ($a \neq b$) for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u \in U} s(u) \leq 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{12}$$

3. *Disjoint k-element sets $A, B \subseteq V$ satisfy $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{a \in A} s(a) = \sum_{b \in B} s(b) \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{13}$$

Proof. For any $c \in \mathcal{C}_d(H)$, let $s_j = \mathbf{1}_{c^{-1}(\{j\})} \in \mathcal{S}_{\text{ext}}(H)$ ($0 \leq j \leq d-1$) be the indicator functions of the d columns of the associated $n \times d$ Context–Color Matrix M. For any subset $X \subseteq V$ and color $j \in \{0, \ldots, d-1\}$, the column sum

$$\sum_{x \in X} s_j(x) = \big|\{x \in X : c(x) = j\}\big| \tag{14}$$

counts the exact multiplicity of color j on X in coloring c. (1) When $|U| = d$, every color $j \in \{0, \ldots, d-1\}$ has multiplicity 1 on U iff $\{c(u) : u \in U\} = \{0, \ldots, d-1\}$, establishing Eq. (11). (2) Every color j has multiplicity at most 1 on U iff no two elements of U share a color in c, establishing Eq. (12). (3) By Eq. (14), color j has the same multiplicity on A and on B for every $j \in \{0, \ldots, d-1\}$ and every $c \in \mathcal{C}_d(H)$ iff $\sum_{a \in A} s_j(a) = \sum_{b \in B} s_j(b)$ for every $s_j \in \mathcal{S}_{\text{ext}}(H)$, establishing Eq. (13). □

Corollary 4 (Affine Pseudocontexts Imply Color-Multiset Equivalence)**.** *Let $M_H \in \{0,1\}^{n \times |V|}$ be the context-vertex incidence matrix of H. If disjoint nonorthogonal*

k-tuples $A, B \subseteq V$ admit a balanced context certificate $\lambda \in \mathbb{R}^n$ satisfying [2]

$$M_H^{\mathsf{T}}\lambda = \chi_A - \chi_B, \qquad \mathbf{1}_n^{\mathsf{T}}\lambda = 0, \tag{15}$$

then in every strong d-coloring $c \in \mathcal{C}_d(H)$, A and B carry the exact same multiset of colors: $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$.

Proof. Every extendable state $s \in \mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$ satisfies $M_H s = \mathbf{1}_n$, so

$$\sum_{a\in A} s(a) - \sum_{b\in B} s(b) = (\chi_A - \chi_B)^{\mathsf{T}} s = \lambda^{\mathsf{T}} M_H s = \lambda^{\mathsf{T}} \mathbf{1}_n = 0.$$

Applying Theorem 3(3) completes the proof. □

III. PAIRED AFFINE-FORCED PSEUDOCONTEXTS IN $\mathbb{R}^3$ AND $\mathbb{C}^3$

A. Mayet's 15-Ray, 8-Context Real Configuration $M_{15,8}$

Consider the 15-vertex, 8-context hypergraph $M_{15,8}$ studied in Refs. [1, 2, 12, 13], with contexts

$$\begin{aligned} C_1 &= \{1,2,3\}, & C_2 &= \{3,4,5\}, \\ C_3 &= \{6,7,8\}, & C_4 &= \{8,9,10\}, \\ C_5 &= \{11,12,13\}, & C_6 &= \{13,14,15\}, \\ C_7 &= \{2,7,12\}, & C_8 &= \{4,9,14\}. \end{aligned} \tag{16}$$

Clearing denominators in Eqs. (9) of Ref. [1] supplies an exact integer FOR in $\mathbb{R}^3$ (zero extra orthogonalities among all $\binom{15}{2} = 105$ pairs):

$$\begin{aligned} r_1 &= (5,3,-4), & r_2 &= (5,-7,1), \\ r_3 &= (1,1,2), & r_4 &= (1,1,-1), \\ r_5 &= (1,-1,0), & r_6 &= (5,-16,13), \\ r_7 &= (5,4,3), & r_8 &= (-2,1,2), \\ r_9 &= (1,0,1), & r_{10} &= (1,4,-1), \\ r_{11} &= (-5,18,1), & r_{12} &= (5,2,-11), \\ r_{13} &= (4,1,2), & r_{14} &= (-1,2,1), \\ r_{15} &= (1,2,-3). \end{aligned} \tag{17}$$

The two primary target triples $A = \{1,6,11\}$ and $B = \{5,10,15\}$ are internally pairwise nonorthogonal. The two 4-context coverings $\mathcal{F}_+ = \{C_1, C_3, C_5, C_8\}$ and $\mathcal{F}_- = \{C_2, C_4, C_6, C_7\}$ partition $V \setminus B$ and $V \setminus A$, respectively, giving $\sum_{C\in\mathcal{F}_+} \chi_C - \sum_{C\in\mathcal{F}_-} \chi_C = \chi_A - \chi_B$. In addition, omitting any one of the three horizontal 2-context chains (C_1, C_2), (C_3, C_4), or (C_5, C_6) leaves a 6-context hexagon whose 3-context coverings force three further paired size-three affine pseudocontexts: $\{1,6,14\}$ vs. $\{5,10,12\}$, $\{1,9,11\}$ vs. $\{5,7,15\}$, and $\{4,6,11\}$ vs. $\{2,10,15\}$ (4 pairs in total).

Exhaustive enumeration gives 24 two-valued states of $M_{15,8}$, all extendable, and $N_c = 16$ normalized 8×3 Context–Color Matrices (with anchor $C_1 = (1,2,3)$). By Corollary 4, the two target triples have identical color multisets in every coloring:

- In 4 normalized colorings, both A and B are *rainbow* $\{0,1,2\}$.
- In 12 normalized colorings, both A and B carry a *two-color multiset*: 4 with $\{0,0,1\}$, 4 with $\{0,1,1\}$, and 4 with $\{0,0,2\}$. (Neither triple is ever monochromatic because $p(1)+p(6)+p(11) \le 2$ on $\mathcal{S}(M_{15,8})$ [1].)

Two explicit 8×3 Context–Color Matrices illustrating the rainbow regime ($M_1^{(M_{15,8})}$) and the two-color multiset regime ($M_2^{(M_{15,8})}$) are:

$$M_1^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c}_0 & \mathbf{c}_1 & \mathbf{c}_2 \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ 8 & \mathbf{6} & 7 \\ 8 & 9 & \mathbf{10} \\ 12 & 13 & \mathbf{11} \\ \mathbf{15} & 13 & 14 \\ \hline 12 & 2 & 7 \\ 4 & 9 & 14 \end{array}\right), \quad M_2^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c}_0 & \mathbf{c}_1 & \mathbf{c}_2 \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ \mathbf{6} & 8 & 7 \\ \mathbf{10} & 8 & 9 \\ 12 & \mathbf{11} & 13 \\ \mathbf{15} & 14 & 13 \\ \hline 12 & 2 & 7 \\ 4 & 14 & 9 \end{array}\right). \tag{18}$$

In $M_1^{(M_{15,8})}$, both $A = \{1,6,11\}$ and $B = \{5,10,15\}$ span all three columns $(\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2)$ via $(1,6,11)$ and $(15,5,10)$. In $M_2^{(M_{15,8})}$, both A and B have two entries in Column $\mathbf{c}_0$ ($\{1,6\}$ and $\{10,15\}$), one entry in Column $\mathbf{c}_1$ (11 and 5), and none in Column $\mathbf{c}_2$. Figure 1 displays both colorings on the Greechie diagram of $M_{15,8}$.

B. Balanced Size-Three Pseudocontexts in $B_{13,7}$ and $R_{21,12}, R_{25,15}$

Two further families of faithfully coordinatized hypergraphs from Ref. [9] already contain balanced covering certificates for size-three affine pseudocontexts:

1. *The 13-ray, 7-context Specker bug $B_{13,7} \subset \mathbb{R}^3$ ($N_c = 6$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 12/14$):* Its 7 contexts are $E_1 = \{1,2,3\}$, $E_2 = \{7,8,9\}$, $E_3 = \{2,8,5\}$, $E_4 = \{3,11,14\}$, $E_5 = \{7,11,15\}$, $E_6 = \{9,10,23\}$, and $E_7 = \{1,10,19\}$ [Fig. 4(a)], with Yu–Oh coordinates from Table I. The six outer contexts partition into two 3-context coverings $\mathcal{F}_+ = \{E_1, E_5, E_6\}$ and $\mathcal{F}_- = \{E_2, E_4, E_7\}$ of $(V \setminus \{5\}) \setminus B$ and $(V \setminus \{5\}) \setminus A$, where

$$A = \{2,15,23\}, \qquad B = \{8,14,19\}. \tag{19}$$

Hence $(\chi_{E_1} + \chi_{E_5} + \chi_{E_6}) - (\chi_{E_2} + \chi_{E_4} + \chi_{E_7}) = \chi_A - \chi_B$, forcing $\{\{c(2), c(15), c(23)\}\} = \{\{c(8), c(14), c(19)\}\}$ across all $N_c = 6$ normalized 7×3 Context–Color Matrices (and likewise in the 4-dimensional common-apex lift $B_{14,7}^{(4)}$).

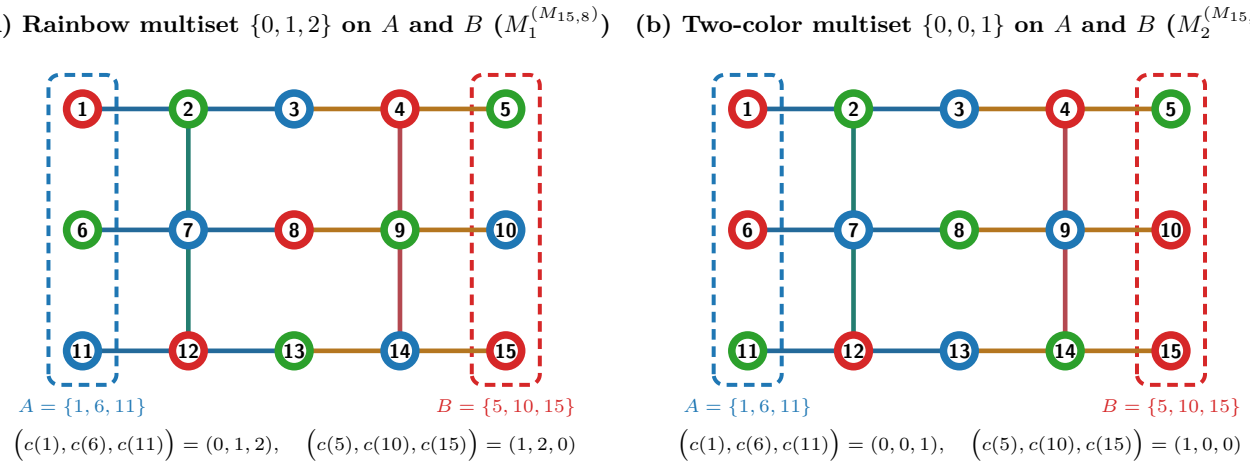


FIG. 1. Mayet's 15-ray, 8-context hypergraph $M_{15,8}$ in $\mathbb{R}^3$ [Eq. (17)] with paired size-three affine pseudocontexts $A = \{1, 6, 11\}$ and $B = \{5, 10, 15\}$ (dashed boxes). Vertex outer rings indicate colors 0 (red), 1 (green), and 2 (blue) for the two Context–Color Matrices in Eq. (18): (a) $M_1^{(M_{15,8})}$, where both A and B are rainbow $\{0, 1, 2\}$; (b) $M_2^{(M_{15,8})}$, where both A and B carry the two-color multiset $\{0, 0, 1\}$.

2. *Gaussian-integer row–column hub grids $R_{21,12}$ and $R_{25,15}$ in $\mathbb{C}^3$ [9, Sec. IV D]:* In $R_{21,12}$ ($N_c = 32$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/54$) and $R_{25,15}$ ($N_c = 24$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/81$), the 3×3 core grid has row contexts $R_1 = \{1, 2, 3\}$, $R_2 = \{4, 5, 6\}$, $R_3 = \{7, 8, 9\}$ and hub contexts $\{1, 10, 13\}$, $\{2, 11, 14\}$, $\{3, 12, 15\}$ (for R_1), $\{4, 10, 16\}$, $\{5, 11, 17\}$, $\{6, 12, 18\}$ (for R_2), and $\{7, 10, 19\}$, $\{8, 11, 20\}$, $\{9, 12, 21\}$ (for R_3). For any two rows (say R_1 and R_2) and any column $j \in \{1, 2, 3\}$ (say $j = 1$), the two 3-context families $\mathcal{F}_+ = \{\{1, 2, 3\}, \{4, 10, 16\}, \{2, 11, 14\}, \{3, 12, 15\}\} \setminus \{\{1, 2, 3\}\}$ combine to yield 12 size-three affine pseudocontexts such as

$$A = \{1, 17, 18\}, \qquad B = \{4, 14, 15\}, \tag{20}$$

via the balanced 3-context identity $(\chi_{\{1,2,3\}} + \chi_{\{5,11,17\}} + \chi_{\{6,12,18\}}) - (\chi_{\{4,5,6\}} + \chi_{\{2,11,14\}} + \chi_{\{3,12,15\}}) = \chi_{\{1,17,18\}} - \chi_{\{4,14,15\}}$, forcing $\{\{c(1), c(17), c(18)\}\} = \{\{c(4), c(14), c(15)\}\}$ in all 32 (resp. 24) normalized Context–Color Matrices.

C. The 20-Ray, 12-Context Complex Octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$

To reach target size $k = 2$ via a balanced covering certificate, one must pass from $\mathbb{R}^3$ to $\mathbb{C}^3$ [2, 5]. The 20-ray, 12-context octagon H_{20} (Fig. 2) has eight perimeter contexts $E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}$ ($j \in \mathbb{Z}_8$) and four inner contexts $D_0 = \{v_0, a_1, v_4\}$, $D_2 = \{v_2, a_2, v_6\}$, $D_1 = \{v_1, b_2, v_5\}$, $D_3 = \{v_3, b_1, v_7\}$. The two 6-context coverings $\mathcal{G}_+ = \{D_0, D_2, E_1, E_3, E_5, E_7\}$ and $\mathcal{G}_- = \{D_1, D_3, E_0, E_2, E_4, E_6\}$ yield $\sum_{C \in \mathcal{G}_+} \chi_C - \sum_{C \in \mathcal{G}_-} \chi_C = \chi_A - \chi_B$ for the nonorthogonal pairs

$$A = \{a_1, a_2\}, \qquad B = \{b_1, b_2\}. \tag{21}$$

Exhaustive computation gives $|\mathcal{S}(\mathsf{H}_{20})| = |\mathcal{S}_{\text{ext}}(\mathsf{H}_{20})| = 26$ two-valued states and $N_c = 12$ normalized 12×3 Context–Color Matrices (with anchor $E_0 = (v_{70}, v_0, v_{01})$). Across all 12 normalized colorings, $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ holds identically: 4 colorings assign monochromatic multisets (2 with $\{0, 0\}$ and 2 with $\{2, 2\}$) and 8 assign bichromatic multisets (2 with $\{0, 1\}$, 4 with $\{0, 2\}$, and 2 with $\{1, 2\}$). Representative bichromatic ($M_1^{(\mathsf{H}_{20})}$) and monochromatic ($M_2^{(\mathsf{H}_{20})}$) Context–Color Matrices are:

$$M_1^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_1 & v_{12} & v_{01} \\ v_{23} & v_{12} & v_2 \\ v_{23} & v_{34} & v_3 \\ v_4 & v_{34} & v_{45} \\ v_{56} & v_5 & v_{45} \\ v_{56} & v_6 & v_{67} \\ v_{70} & v_7 & v_{67} \\ \hline v_4 & v_0 & \mathbf{a_1} \\ \mathbf{a_2} & v_6 & v_2 \\ v_1 & v_5 & \mathbf{b_2} \\ \mathbf{b_1} & v_7 & v_3 \end{array}\right), \quad M_2^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_{12} & v_1 & v_{01} \\ v_{12} & v_2 & v_{23} \\ v_{34} & v_3 & v_{23} \\ v_{34} & v_{45} & v_4 \\ v_{56} & v_{45} & v_5 \\ v_{56} & v_{67} & v_6 \\ v_{70} & v_{67} & v_7 \\ \hline \mathbf{a_1} & v_0 & v_4 \\ \mathbf{a_2} & v_2 & v_6 \\ \mathbf{b_2} & v_1 & v_5 \\ \mathbf{b_1} & v_3 & v_7 \end{array}\right). \tag{22}$$

IV. A TWO-VALUED STATE ADMISSIBILITY CONTEXT IN $\mathbb{R}^3$

We first realize the stronger alternative in Definition 1: an independent triple normalized on *every* two-valued state. The construction pastes two standard nonseparating gadgets at one ray. Its logical proof is independent of

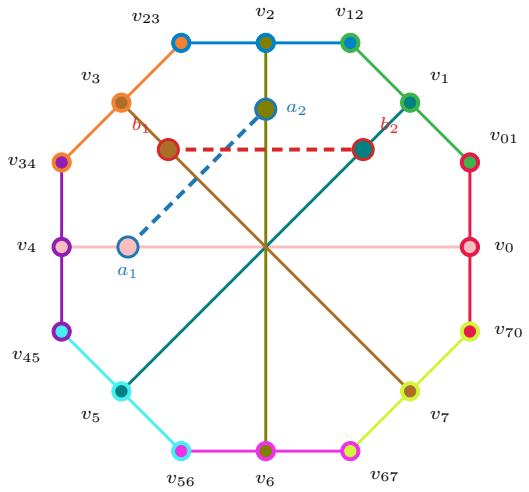


FIG. 2. The 20-ray, 12-context complex octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$ [2, 5]. The two nonorthogonal target pairs $A = \{a_1, a_2\}$ (blue dashed link) and $B = \{b_1, b_2\}$ (red dashed link) carry identical 2-element color multisets $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ across all $N_c = 12$ normalized 12×3 Context–Color Matrices [Eq. (22)].

the subsequent enumeration, and an explicit FOR makes the distinction from affine and operator normalization concrete.

A. A Completed Nonseparating Gadget

Write $\mathcal{B}(a, b; x)$ for the seven-context Specker bug

$$\begin{gathered} \{a, x_1, x_2\}, \quad \{x_2, x_3, x_4\}, \\ \{x_4, x_5, b\}, \quad \{b, x_6, x_7\}, \\ \{x_7, x_8, x_9\}, \quad \{x_9, x_{10}, a\}, \\ \{x_3, x_8, x_{11}\}. \end{gathered} \tag{23}$$

Every two-valued state satisfies $s(a) + s(b) \leq 1$. Indeed, assigning one to both endpoints forces $s(x_2) = s(x_4) = s(x_7) = s(x_9) = 0$, hence $s(x_3) = s(x_8) = 1$, contradicting the final context.

Let G be the 27-ray, 17-context gadget formed by

$$\begin{gathered} \mathcal{B}(a, b; x), \quad \mathcal{B}(b', a'; y), \\ \{a, c, b'\}, \quad \{b, c, a'\}, \quad \{c, x_{11}, y_{11}\}. \end{gathered} \tag{24}$$

This is a completed realization of the Kochen–Specker Γ_3 nonseparating construction [14, 15]. The last context completes an orthogonal triple in the FOR below; the logical identity already follows from the other 16 contexts. In Fig. 3, the labels E_{15}, E_{16}, E_{17} follow the order of these three additional contexts; E_{32}, E_{33}, E_{34} label the corresponding contexts of the second copy. Each repeated label pairs the two branches of one context through its intertwining vertex.

Lemma 5 (Two-Valued Nonseparation)**.** *Every $s \in \mathcal{S}(G)$ satisfies*

$$\begin{gathered} s(a) = s(a'), \qquad s(b) = s(b'), \\ s(a) + s(b) + s(c) = 1. \end{gathered} \tag{25}$$

Proof. If $s(a) = 1$, the first bug forces $s(b) = 0$, and $\{a, c, b'\}$ forces $s(c) = s(b') = 0$. The context $\{b, c, a'\}$ then gives $s(a') = 1$. Conversely, if $s(a') = 1$, the second bug forces $s(b') = 0$, while $\{b, c, a'\}$ forces $s(b) = s(c) = 0$; now $\{a, c, b'\}$ gives $s(a) = 1$. Thus $s(a) = s(a')$. The symmetric argument gives $s(b) = s(b')$. Substitution into $s(a) + s(c) + s(b') = 1$ proves the final identity. □

B. One-Vertex Pasting and Universal Admissibility

Take a second copy $\widetilde{G}$ and identify only its $\widetilde{a}$ with the first copy's c. Write $u = \widetilde{a}'$. The union $H_{53,34} = G \cup \widetilde{G}$ has $27 + 27 - 1 = 53$ rays and $17 + 17 = 34$ contexts, all drawn in Fig. 3.

Theorem 6 (Two-Valued State Admissibility on Nonadjacent Rays)**.** *The hypergraph $H_{53,34}$ admits a FOR in $\mathbb{R}^3$ in which $U = \{a, b, u\}$ is pairwise nonorthogonal. Its two-valued state space is nonempty and*

$$s(a) + s(b) + s(u) = 1 \qquad \forall s \in \mathcal{S}(H_{53,34}). \tag{26}$$

Thus U is a two-valued state admissibility context and, in particular, a chromatic admissible context.

Proof. Restrict any two-valued state to each copy of G. Lemma 5 gives $s(a) + s(b) + s(c) = 1$ in the first copy and $s(c) = s(u)$ in the second, proving Eq. (26). The representation below verifies pairwise nonorthogonality and faithfulness. The enumeration in Eq. (34) establishes nonemptiness and shows that each distinguished vertex can be true. Every color-class state of every strong 3-coloring is a two-valued state, so Theorem 3(1) gives universal chromatic completeness as well. □

C. An Explicit Faithful Real Representation

Use the unnormalized ray representatives

$$\begin{aligned} q_a &= (1, -1, 1)^{\mathsf{T}}, \\ q_b &= (1, -1, -1)^{\mathsf{T}}, \\ q_c &= (1, 1, 0)^{\mathsf{T}}, \end{aligned} \tag{27}$$

and the bug vectors

i	$q_{x_i}^{\mathsf{T}}$	i	$q_{x_i}^{\mathsf{T}}$
1	$(2, 1, -1)$	7	$(1, 0, 1)$
2	$(0, 1, 1)$	8	$(0, 1, 0)$
3	$(1, 0, 0)$	9	$(1, 0, -1)$
4	$(0, 1, -1)$	10	$(1, 2, 1)$
5	$(2, 1, 1)$	11	$(0, 0, 1)$
6	$(1, 2, -1)$		

(28)

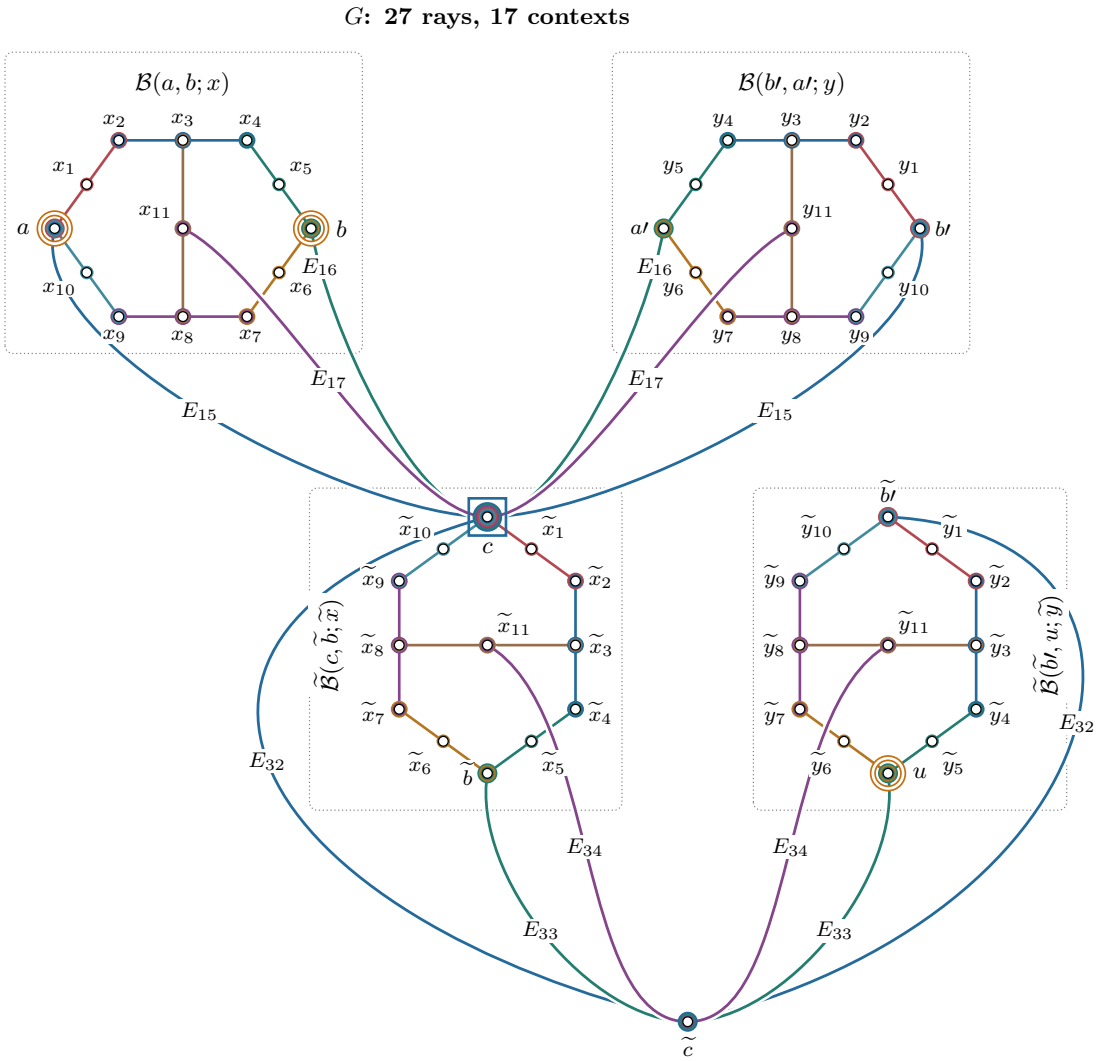


FIG. 3. The hypergraph $H_{53,34}$, with all 53 rays and 34 contexts. Dotted frames delimit seven-context Specker bugs; frame boundaries are not contexts. In G, the connecting contexts are $E_{15}=\{a,c,b'\}$, $E_{16}=\{b,c,a'\}$, and $E_{17}=\{c,x_{11},y_{11}\}$; E_{32},E_{33},E_{34} are their counterparts in $\widetilde{G}$. Repeated labels identify both branches of each connecting context. The curves $E_{15},E_{16},E_{17},E_{33},E_{34}$ are smooth through their middle vertices. The upper bug layouts and E_{15},E_{16},E_{17} are mirror-symmetric about the vertical through c; E_{33} and E_{34} are individually symmetric about the vertical through $\widetilde{c}$, where they share a horizontal tangent. Crossings without a vertex symbol are not vertices. The square marks the unique shared ray $c=\widetilde{a}$, giving $27+27-1=53$ rays and $17+17=34$ contexts. Concentric rings record incident contexts; additional double outer outlines mark $U=\{a,b,u\}$, where $u=\widetilde{a'}$. On every two-valued state s, G enforces $s(a)+s(b)+s(c)=1$, while $\widetilde{G}$ enforces $s(c)=s(u)$. Substitution yields $s(a)+s(b)+s(u)=1$, proving two-valued state admissibility of U.

Put $n=q_c/\sqrt{2}$ and $R=nn^{\mathsf{T}}+[n]_\times$, where $[n]_\times v=n\times v$. This is a 90° rotation about n. The remaining rays of G are

$$q_{b'}=Rq_a,\quad q_{a'}=Rq_b,\quad q_{y_i}=Rq_{x_i}. \tag{29}$$

For the second copy, set $Q=L_0E_0F_0^{\mathsf{T}}$, with

$$\begin{aligned} F_0&=\begin{pmatrix}\dfrac{q_a}{\sqrt{3}} & \dfrac{(1,-1,-2)^{\mathsf{T}}}{\sqrt{6}} & n\end{pmatrix},\\ E_0&=\begin{pmatrix}n & (0,0,1)^{\mathsf{T}} & \dfrac{(1,-1,0)^{\mathsf{T}}}{\sqrt{2}}\end{pmatrix},\\ L_0&=\frac{1}{11}\begin{pmatrix}9&2&6\\2&9&-6\\-6&6&7\end{pmatrix}. \end{aligned} \tag{30}$$

The columns of E_0, F_0 are orthonormal, and L_0 is a rotation fixing n. Consequently, Qq_a is parallel to q_c, as required for the identified vertex. Assign Qq_v to every other ray of the second copy. In particular,

$$Qq_{a'} = -\frac{1}{11\sqrt{3}}(16, 28, -7)^{\mathsf{T}}. \tag{31}$$

Choosing $q_u = (16, 28, -7)^{\mathsf{T}}$ gives

$$q_a \cdot q_b = 1, \quad q_a \cdot q_u = -19, \quad q_b \cdot q_u = -5, \tag{32}$$

so the target triple is pairwise nonorthogonal. Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ verifies the full FOR: among all $\binom{53}{2} = 1378$ pairs, precisely the 102 pairs belonging to the 34 contexts are orthogonal, and no two rays are parallel. The accompanying certificate lists all rays, contexts, states, and normalized colorings, and includes a reproducible verifier.

D. State Counts and Separation from Operator Normalization

Exhaustive enumeration gives

$$\begin{aligned} &|\mathcal{S}(H_{53,34})| = 224, \qquad |\mathcal{S}_{\text{ext}}(H_{53,34})| = 128, \\ &N_c = 128 \quad \text{(768 labeled strong 3-colorings)}. \end{aligned} \tag{33}$$

The restrictions of all two-valued states to U are

$(s(a), s(b), s(u))$	multiplicity
$(1, 0, 0)$	80
$(0, 1, 0)$	80
$(0, 0, 1)$	64

(34)

There are exactly two independent chromatic admissible triples, $\{a, b, u\}$ and $\{a', b', u\}$; both are normalized on every two-valued state. Table II records the remaining chromatic census counts for this configuration. The total of 25 non-context complete rainbow triples consists of these two independent triples and 23 partially nonorthogonal triples. No minimality claim is made for the 53 rays or 34 contexts.

The normalization in Eq. (26) has no affine certificate. The context incidence matrix has rank 34, and adjoining the row χ_U^{T} raises its rank to 35. The geometric separation is also explicit: for the Born preparation along q_c,

$$\begin{aligned} &p_c(a) = p_c(b) = 0, \\ &p_c(u) = \frac{|q_c \cdot q_u|^2}{\|q_c\|^2 \|q_u\|^2} = \frac{44^2}{2 \cdot 33^2} = \frac{8}{9}. \end{aligned} \tag{35}$$

Thus every classical two-valued state, and every convex mixture of them, gives total one on U, while this quantum state gives $8/9$. Two-valued state admissibility therefore does not imply an affine or operator normalization identity.

V. CHROMATIC ADMISSIBLE CONTEXTS AND COLOR-MULTISET EQUIVALENCE IN $\mathbb{R}^3$

A. Chromatic Admissibility and Operator Rigidity

Before exhibiting chromatic admissible contexts in $\mathbb{R}^3$, we distinguish their state-restricted normalization from an affine or operator identity.

Proposition 7 (Operator and Affine Rigidity of Complete d-Tuples)**.** *Let $H = (V, E)$ be a d-uniform hypergraph admitting a FOR $v \mapsto q_v \in \mathbb{C}^d$ with rank-one projectors $P_v = q_v q_v^\dagger / \|q_v\|^2$. If a d-element subset $U = \{u_1, \ldots, u_d\} \subseteq V$ satisfies $\sum_{u \in U} P_u = I_d$ (in particular, if $\chi_U = M_H^{\mathsf{T}} \lambda$ with $\mathbf{1}_n^{\mathsf{T}} \lambda = 1$), then $q_{u_a} \perp q_{u_b}$ for all $1 \le a < b \le d$. Consequently, no set of d nonorthogonal rays can sum to I_d or be forced to sum to 1 by a linear combination of context equations.*

Proof. For each $a \in \{1, \ldots, d\}$, conjugating $I_d - P_{u_a} = \sum_{b \neq a} P_{u_b}$ by the unit vector $\hat{q}_{u_a} = q_{u_a} / \|q_{u_a}\|$ gives

$$0 = \langle \hat{q}_{u_a}, (I_d - P_{u_a}) \hat{q}_{u_a} \rangle = \sum_{b \neq a} \left| \langle \hat{q}_{u_a}, \hat{q}_{u_b} \rangle \right|^2,$$

so $\langle q_{u_a}, q_{u_b} \rangle = 0$ for all $b \neq a$. If $\chi_U = M_H^{\mathsf{T}} \lambda$ with $\mathbf{1}_n^{\mathsf{T}} \lambda = 1$, summing $\sum_{v \in C_i} P_v = I_d$ with weights λ_i yields $\sum_{u \in U} P_u = (\mathbf{1}_n^{\mathsf{T}} \lambda) I_d = I_d$, forcing mutual orthogonality. □

Proposition 7 excludes operator completeness and an affine normalization certificate for any d-tuple containing a nonorthogonal pair. It does not exclude normalization on a specified family of two-valued states. In particular, chromatic admissibility imposes Eq. (11) only on $\mathcal{S}_{\text{ext}}(H)$. Whether this identity extends to all of $\mathcal{S}(H)$ is a separate question; the proposition alone does not imply that a violating nonextendable state must exist. The construction of Section IV supplies an explicit example normalized on all two-valued states but not on all Born states.

A sufficient additional hypothesis for lifting every identity on two-valued states to an affine identity is

$$\operatorname{aff} \mathcal{S}(H) = \mathcal{A}(H). \tag{36}$$

Under this hypothesis, normalization on all of $\mathcal{S}(H)$ extends to $\mathcal{A}(H)$ and hence has a context-equation certificate. Consequently, in a faithful orthogonal representation, an independent chromatic admissible context must then violate normalization on some nonextendable two-valued state. Without Eq. (36), no such general necessity is asserted. The examples below distinguish the two state families directly through their enumerated state sums.

B. Non-Twin Color-Multiset-Equivalent Pairs of Size Two in $\mathbb{R}^3$ ($I_{16,9}, H_{17,10}, T_{19,11}$)

A faithful real three-dimensional hypergraph can support a non-twin color-multiset-equivalent pair of size $k = 2$

TABLE I. Unnormalized integer vectors in $\mathbb{R}^3$ for the 25-ray completed Yu–Oh configuration H^+_{YO} (rays $1,\dots,25$, used by $B_{13,7}$, $H_{17,10}$, and $H_{21,13}$) and for the 19-ray gadget $T_{19,11}$ (second tuple where different, plus tip rays $26, 27, 28$) [8, 16]. For $H_{20,12}$ (omitting $\{6, 13, 22\}$), taking $r_1 = (0, 1, 2)$, $r_3 = (0, 2, -1)$, $r_7 = (2, 0, 1)$, $r_9 = (1, 0, -2)$, $r_{10} = (2, -2, 1)$, $r_{11} = (1, -1, -2)$, $r_{13} = (1, 4, -2)$ and cross-product completions yields a strictly faithful integer representation with $\langle r_6, r_{13}\rangle = -3 \neq 0$.

i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)
1	$(0,1,1)$ / $(0,1,2)$	8	$(0,1,0)$	15	$(1,2,-1)$ / $(1,4,-1)$	22	$(1,1,2)$
2	$(1,0,0)$	9	$(1,0,-1)$	16	$(1,-2,1)$	23	$(1,2,1)$ / $(1,4,1)$
3	$(0,1,-1)$ / $(0,2,-1)$	10	$(1,-1,1)$ / $(2,-1,2)$	17	$(1,1,-2)$	24	$(1,-1,2)$ / $(4,-2,5)$
4	$(1,1,0)$ / $(1,2,0)$	11	$(1,-1,-1)$ / $(2,-1,-2)$	18	$(1,-1,-2)$ / $(4,-2,-5)$	25	$(2,-1,-1)$
5	$(0,0,1)$	12	$(1,1,1)$	19	$(2,1,-1)$	26	$(2,-41,18)$ ($T_{19,11}$)
6	$(1,-1,0)$ / $(2,-1,0)$	13	$(1,1,-1)$	20	$(2,-1,1)$	27	$(200,-82,-209)$ ($T_{19,11}$)
7	$(1,0,1)$	14	$(2,1,1)$ / $(5,2,4)$	21	$(1,-2,-1)$	28	$(241,82,160)$ ($T_{19,11}$)

without an affine certificate. This does not contradict the prohibition of size-two real affine pseudocontexts in Ref. [2]: the color relation is required only on $\mathcal{S}_{\mathrm{ext}}(H)$. The construction combines a linear context identity with a non-context complete rainbow triple.

Theorem 8 (Non-Twin Color-Multiset Equivalence of Size Two in $\mathbb{R}^3$)**.** *In the faithful real three-dimensional hypergraphs $I_{16,9}$ (16 rays, 9 contexts, $N_c = 6$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 12/22$) and $H_{17,10}$ (17 rays, 10 contexts, $N_c = 4$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 10/18$), the nonorthogonal pairs*

$$\begin{aligned} A_1 &= \{2,4\}, \quad B_1 = \{14,19\}, \\ A_2 &= \{4,8\}, \quad B_2 = \{15,23\} \end{aligned} \tag{37}$$

have no chromatic twins between A_r and B_r, yet satisfy

$$\begin{aligned} \{\{c(2),c(4)\}\} &= \{\{c(14),c(19)\}\}, \\ \{\{c(4),c(8)\}\} &= \{\{c(15),c(23)\}\} \end{aligned} \qquad \forall\, c \in \mathcal{C}_3. \tag{38}$$

Similarly, in $T_{19,11}$ (19 rays, 11 contexts, $N_c = 7$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 14/25$, Fig. 4(b)), the partially nonorthogonal tip triple $U_{\mathrm{tip}} = \{26, 27, 28\}$ (with $26 \perp 27$, $26 \perp 28$, $27 \not\perp 28$) is rainbow in every 3-coloring, and the nonorthogonal pairs $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$ satisfy $\{\{c(3), c(10)\}\} = \{\{c(24), c(26)\}\}$ for all $c \in \mathcal{C}_3(T_{19,11})$.

Proof. Both $I_{16,9}$ and $H_{17,10}$ contain the Specker-bug contexts $E_1 = \{1,2,3\}$, $E_2 = \{7,8,9\}$, $E_4 = \{2,8,5\}$, $E_8 = \{3,11,14\}$, $E_9 = \{7,11,15\}$, $E_{11} = \{9,10,23\}$, $E_{12} = \{1,10,19\}$ together with the two roof contexts $E_{10} = \{4,11,24\}$ and $E_{13} = \{4,10,18\}$ ($H_{17,10}$ also includes $E_3 = \{4,5,6\}$). On the Specker bug, every 3-coloring satisfies the universally rainbow condition $c(10) \neq c(11)$. Since 4 is adjacent to 11 in E_{10} and to 10 in E_{13}, $c(4) \neq c(11)$ and $c(4) \neq c(10)$. Thus $\{4, 10, 11\}$ is a partially nonorthogonal complete rainbow triple: $\{c(4), c(10), c(11)\} = \{0, 1, 2\}$ for all $c \in \mathcal{C}_3$, or equivalently by Theorem 3(1),

$$s(4) + s(10) + s(11) = 1 \qquad \forall\, s \in \mathcal{S}_{\mathrm{ext}}. \tag{39}$$

Meanwhile, on *every* two-valued state $s \in \mathcal{S}$, the context combination $E_8 + E_{12} - E_1$ gives

$$\begin{aligned} \big(s(3) + s(11) + s(14)\big) + \big(s(1) + s(10) + s(19)\big) \\ - \big(s(1) + s(2) + s(3)\big) = 1, \end{aligned} \tag{40}$$

so $s(14) + s(19) - s(2) = 1 - s(10) - s(11)$. Substituting Eq. (39) into Eq. (40) yields $s(2) + s(4) = s(14) + s(19)$ for all $s \in \mathcal{S}_{\mathrm{ext}}$, and symmetric combination $E_9 + E_{11} - E_2$ yields $s(4) + s(8) = s(15) + s(23)$ for all $s \in \mathcal{S}_{\mathrm{ext}}$. By Theorem 3(3), Eq. (38) holds in every 3-coloring. In $H_{17,10}$ [Eq. (14) of Ref. [9]], $(c(2), c(4))$ and $(c(14), c(19))$ equal $(0, 0)$ and $(0, 0)$ in c_1, c_3, $(0, 1)$ and $(1, 0)$ in c_2, and $(0, 1)$ and $(0, 1)$ in c_4, confirming the absence of chromatic twins between A_1 and B_1.

In $T_{19,11}$ [Fig. 4(b)], the two upper contexts $\{14, 27, 26\}$ and $\{18, 28, 26\}$ together with the universally rainbow pair $c(14) \neq c(18)$ force $18 \sim_c 27$, $14 \sim_c 28$, and $\{c(26), c(27), c(28)\} = \{0, 1, 2\}$. Combining $\{3, 11, 14\} + \{4, 10, 18\} - \{4, 11, 24\}$ with $c(18) = c(27)$ and $\{14, 27, 26\}$ gives $s(3) + s(10) = s(24) + s(26)$ on $\mathcal{S}_{\mathrm{ext}}(T_{19,11})$. □

C. Chromatic Admissible Contexts in $H_{20,12}$ and $H_{21,13}$

Finally, we turn to Class I of Definition 2: a single triple $U = \{u_1, u_2, u_3\}$ of *pairwise nonorthogonal* rays in $\mathbb{R}^3$ ($q_{u_a} \not\perp q_{u_b}$ for all $a \neq b$) that is chromatically complete ($\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$) in *every* strong 3-coloring $c \in \mathcal{C}_3(H)$.

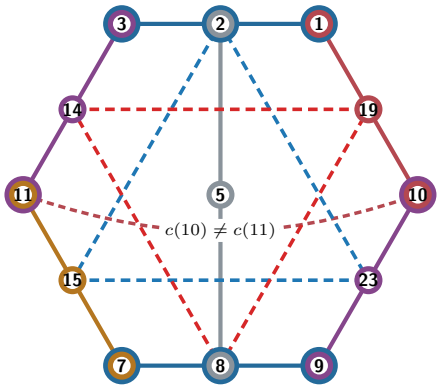
Lemma 9 (Structural Color Propagation in $H_{21,13}$)**.** *Every strong 3-coloring gives distinct colors to the hubs $10, 11, 13$ and to each of the triples $\{1, 4, 7\}$ and $\{3, 6, 9\}$. There are exactly two normalized strong 3-colorings, representing all 12 labeled colorings.*

Proof. Suppose first that hubs $10, 11$ share color t. Vertices $1, 3$ both avoid t, so their context forces $c(2) = t$. Similarly, vertices $7, 9$ both avoid t, forcing $c(8) = t$, contrary to the context $\{2, 5, 8\}$. Equating hubs $10, 13$ similarly forces $c(5) = c(8)$, while equating $11, 13$ forces $c(2) = c(5)$. Thus the hub triple is rainbow.

Adjacency to two hubs now forces

$$c(1) = c(11), \quad c(4) = c(13), \quad c(7) = c(10),$$

(a) Specker bug $B_{13,7}$: $c(10) \neq c(11)$ and $A = \{2, 15, 23\}$, $B = \{8, 14, 19\}$



(b) $T_{19,11}$: rainbow $\{26, 27, 28\}$ and size-two $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$

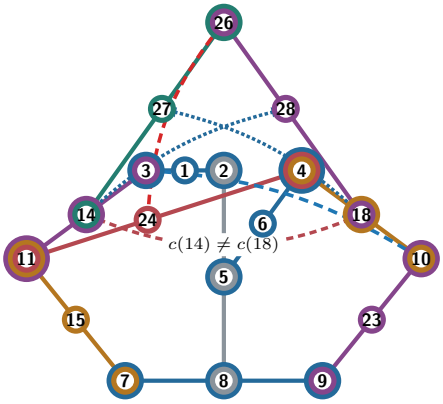


FIG. 4. Three-dimensional gadgets in $\mathbb{R}^3$ (coordinates in Table I): (a) Specker bug $B_{13,7}$ with universally rainbow pair $c(10) \neq c(11)$ and size-three affine pseudocontexts $A = \{2, 15, 23\}$ (blue dashed triangle) and $B = \{8, 14, 19\}$ (red dashed triangle); (b) $T_{19,11}$ with universally rainbow pair $c(14) \neq c(18)$, chromatic twins $14 \sim_c 28$ and $18 \sim_c 27$, complete rainbow tip triple $\{26, 27, 28\}$, and the non-twin color-multiset-equivalent pair $A_3 = \{3, 10\}$ (blue dashed arc) and $B_3 = \{24, 26\}$ (red dashed arc).

so $\{1, 4, 7\}$ is rainbow. Arrange the three base contexts as rows of

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Every row contains each color once. The first column is rainbow by the preceding argument, and the middle column is an actual context. Counting occurrences of each color therefore leaves exactly one occurrence in the last column, proving that $\{3, 6, 9\}$ is rainbow.

Normalize the middle column to $(0, 1, 2)$. The first column is a permutation avoiding the middle-column color in each row, so it is one of the two derangements of three colors. The last column, the hubs, and every remaining vertex are then uniquely determined. Both derangements extend to all contexts. Hence there are two normalized colorings, and their six color permutations exhaust all 12 labeled strong colorings. □

Theorem 10 (Sharp Minimality and Transversal Classification in Yu–Oh Restrictions)**.** *Consider the* 13*-context,* 21*-ray Yu–Oh restriction* $H_{21,13}$ *in* $\mathbb{R}^3$ *(Table I and Fig. 5) obtained by omitting the single diagonal sector* $h_0 = 12$ *from* H_{YO}^{+} *[8, 9].*

1. Sharp threshold at 12 contexts ($H_{20,12}$): *Among all* $2^{13} = 8{,}192$ *context subhypergraphs of* $H_{21,13}$*, none with* $|E| \leq 11$ *contexts possesses a pairwise-nonorthogonal complete rainbow triple. At* $|E| = 12$*, deleting a single roof context from one diagonal sector—for instance omitting* $\{6, 13, 22\}$ *to form* $H_{20,12}$ *(*$|V| = 20$*,* $|E| = 12$*,* $N_c = 4$*,* $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 10/25$*)—yields the minimal configuration, which already forces two pairwise-nonorthogonal complete rainbow triples:*

$$U_1^{(20)} = \{15, 18, 20\}, \qquad U_2^{(20)} = \{19, 21, 24\}. \tag{41}$$

2. Even-transversal classification in $H_{21,13}$ ($|V| = 21$, $|E| = 13$, $N_c = 2$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 6/21$): *Each vertex* $v \in V(H_{21,13})$ *has a bivariate color signature* $\sigma(v) = (c_1(v), c_2(v)) \in \{0, 1, 2\}^2$ *across the two normalized* 13×3 *Context–Color Matrices [Eq. (16) of Ref. [9]], partitioning* $V(H_{21,13})$ *into the* 3×3 *signature grid:*

$c_2 \backslash c_1$	$\mathbf{c}_0 = 0$	$\mathbf{c}_1 = 1$	$\mathbf{c}_2 = 2$
0	$\{2, 14\}$	$\{6\}$	$\{7, 10, 20, 24\}$
1	$\{4, 13, 15, 19\}$	$\{8, 23\}$	$\{3\}$
2	$\{9\}$	$\{1, 11, 18, 21\}$	$\{5, 22\}$

(42)

A triple $U \subseteq V(H_{21,13})$ *is rainbow in every* 3*-coloring if and only if* U *is a* permutation transversal *[17] of Eq. (42), selecting one vertex from each row and one from each column (*$\sum_{\pi \in S_3} \prod_{i=0}^{2} |F_{i,\pi(i)}| = 97$ *total transversals, i.e.,* 13 *contexts plus* 84 *non-context rainbow triples). The three odd permutations* $\pi \in S_3$ *contain* 0 *pairwise-nonorthogonal triples, whereas the three even (cyclic) permutations yield* exactly 21 chromatic admissible contexts*:*

- Singleton cyclic transversal $F_{0,2} \times F_{1,0} \times F_{2,1}$ (1 triple):

$$U_{\mathrm{cop}} = \{3, 6, 9\}, \tag{43}$$

whose rays $r_3 = (0, 1, -1)$*,* $r_6 = (1, -1, 0)$*,* $r_9 = (1, 0, -1)$ *are not only pairwise nonorthogonal (*$\langle r_3, r_6 \rangle = -1$*,* $\langle r_3, r_9 \rangle = 1$*,* $\langle r_6, r_9 \rangle = 1$*) but* coplanar *(*$r_3 + r_6 - r_9 = \mathbf{0}$*, spanning the*

plane orthogonal to omitted hub $r_{12} = (1, 1, 1)$ *so* $\operatorname{rank}(P_3 + P_6 + P_9) = 2 < 3$*).*

- Main diagonal transversal $F_{0,0} \times F_{1,1} \times F_{2,2}$ (4 of 8 triples):

$$\{2, 22, 23\}, \quad \{5, 14, 23\}, \quad \{8, 14, 22\}, \quad \{14, 22, 23\}. \tag{44}$$

- Cyclic transversal $F_{0,1} \times F_{1,2} \times F_{2,0}$ (16 of 64 triples):

$$\begin{gathered}\{1, 4, 7\},\ \{1, 15, 24\},\ \{4, 20, 21\},\ \{7, 18, 19\},\\ \{10, 11, 13\},\ \{10, 15, 21\},\ \{11, 19, 20\},\ \{13, 18, 24\},\\ \{15, 18, 20\},\ \{15, 18, 24\},\ \{15, 20, 21\},\ \{15, 21, 24\},\\ \{18, 19, 20\},\ \{18, 19, 24\},\ \{19, 20, 21\},\ \{19, 21, 24\}.\end{gathered} \tag{45}$$

In particular, $U_{\text{core}} = \{1, 4, 7\}$ *consists of three core rays* $r_1 = (0, 1, 1)$, $r_4 = (1, 1, 0)$, $r_7 = (1, 0, 1)$ *at pairwise* $60°$*, and* $U_{\text{hub}} = \{10, 11, 13\}$ *consists of the three retained diagonal hubs* $r_{10} = (1, -1, 1)$, $r_{11} = (1, -1, -1)$, $r_{13} = (1, 1, -1)$.

3. Chromatic twin classes of size four ($k = 4 > d = 3$): *All four rays within each of the three 4-element signature fibers* $F_{0,2} = \{7, 10, 20, 24\}$, $F_{1,0} = \{4, 13, 15, 19\}$, *and* $F_{2,1} = \{1, 11, 18, 21\}$ *of Eq. (42) are pairwise nonorthogonal in* $\mathbb{R}^3$ *and share the exact same color in every 3-coloring of* $H_{21,13}$.

Proof. By Lemma 9, any strong 3-coloring $c \in \mathcal{C}_3(H_{21,13})$ is obtained by applying a color permutation $\tau \in S_3$ to either c_1 or c_2. Hence $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ holds for all $c \in \mathcal{C}_3(H_{21,13})$ iff $\{c_1(u_1), c_1(u_2), c_1(u_3)\} = \{0, 1, 2\}$ and $\{c_2(u_1), c_2(u_2), c_2(u_3)\} = \{0, 1, 2\}$, meaning that the three signatures $\sigma(u_1), \sigma(u_2), \sigma(u_3)$ have distinct first coordinates and distinct second coordinates—the definition of a permutation transversal of Eq. (42). Exact inner-product checks on the 97 transversals give the listed geometric classification and twin classes. Part (1) is a computer-assisted minimality result: all 2^{13} context subhypergraphs were enumerated, their strong colorings computed, and their independent complete triples tested. The elementary universal-completeness and coloring-count arguments are supplied separately by Lemma 9. □

Over all 21 ordinary two-valued states $s \in \mathcal{S}(H_{21,13})$, the sum $s(u_1) + s(u_2) + s(u_3)$ on $U_{\text{core}} = \{1, 4, 7\}$ takes values $0, 1, 2$ with multiplicities $6, 12, 3$, and on $U_{\text{cop}} = \{3, 6, 9\}$ takes values $0, 1, 2$ with multiplicities $3, 12, 6$; on the 6 extendable states $\mathcal{S}_{\text{ext}}(H_{21,13})$ it is identically 1.

D. Two-Valued Admissibility Contexts in $\mathbb{R}^3$

Can an independent triple be normalized on *every* two-valued state of a faithfully represented, 3-colorable hypergraph in $\mathbb{R}^3$? The answer is affirmative. The mechanism below has two separate ingredients: a binary implication

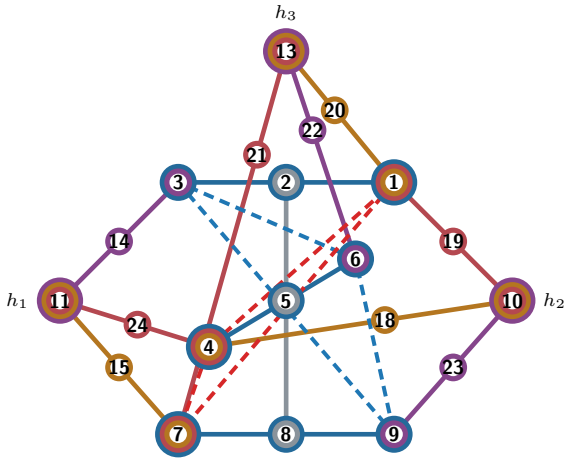


FIG. 5. The 21-ray, 13-context Yu–Oh restriction $H_{21,13}$ in $\mathbb{R}^3$ (all 13 contexts drawn as straight lines; coordinates in Table I). Highlighted are two of the 21 chromatic admissible contexts (Theorem 10): the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ (blue dashed triangle, $r_3 + r_6 - r_9 = \mathbf{0}$) and the $60°$ basis triple $U_{\text{core}} = \{1, 4, 7\}$ (red dashed triangle), both satisfying $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ in every 3-coloring of $H_{21,13}$.

across two contexts, and a geometric realization of the gadgets supplying its exclusivity assumptions. Figure 6 displays the implication in two stages.

Lemma 11 (Roof Twins)**.** *Let* H *contain the contexts* $\{a, x, y'\}$ *and* $\{a, y, x'\}$*. Suppose*

$$s(x) + s(y) \le 1, \qquad s(x') + s(y') \le 1 \tag{46}$$

for every $s \in \mathcal{S}(H)$*. Then every such state satisfies*

$$s(x) = s(x'), \qquad s(y) = s(y'), \tag{47}$$

and consequently

$$\begin{aligned} s(a) + s(x) + s(y) &= 1, \\ s(a) + s(x') + s(y') &= 1. \end{aligned} \tag{48}$$

Proof. Subtracting the two context equations gives

$$\delta = s(x) - s(x') = s(y) - s(y') \in \{-1, 0, 1\}.$$

If $\delta = 1$, then $s(x) = s(y) = 1$; if $\delta = -1$, then $s(x') = s(y') = 1$. Both contradict Eq. (46). Hence $\delta = 0$, proving the twin identities. Substitution into either context equation gives Eq. (48). This step uses the binary range of s. □

Write $B(x, y; \ell)$ for a Specker bug with tips x, y and eleven new vertices $\ell + k$, where $k \in \{1, 2, 3, 5, 7, 8, 9, 14, 15, 19, 23\}$. Its seven contexts are

$$\begin{gathered} \{\ell + 1, \ell + 2, \ell + 3\}, \quad \{\ell + 7, \ell + 8, \ell + 9\}, \\ \{\ell + 2, \ell + 8, \ell + 5\}, \\ \{\ell + 3, y, \ell + 14\}, \quad \{\ell + 7, y, \ell + 15\}, \\ \{\ell + 9, x, \ell + 23\}, \quad \{\ell + 1, x, \ell + 19\}. \end{gathered} \tag{49}$$

These contexts force $s(x)+s(y)\le 1$: if both tips were one, vertices $\ell+1,\ell+3,\ell+7,\ell+9$ would be zero, so the first two contexts would force $s(\ell+2)=s(\ell+8)=1$, contradicting the third context. This is the only bug implication needed for the cascade.

Lemma 12 (Specker-Bug Coordinates)**.** *Let $u,v\in\mathbb{R}^3$ have equal nonzero norm and $0<\langle\hat{u},\hat{v}\rangle\le 1/3$, after reversing one vector if necessary. Set $e_1=u+v$, $e_2=u-v$, $e_3=u\times v$, and*

$$\cos\phi=\sqrt{2-\frac{\|e_1\|^2}{\|e_2\|^2}},\qquad \sin\phi=\sqrt{\frac{\|e_1\|^2}{\|e_2\|^2}-1}. \tag{50}$$

The orthonormal triad

$$\begin{aligned} f_{1,2}&=\tfrac{1}{\sqrt{2}}\big(\hat{e}_1\pm(\cos\phi\,\hat{e}_2+\sin\phi\,\hat{e}_3)\big),\\ f_3&=f_1\times f_2 \end{aligned} \tag{51}$$

gives the intended orthogonalities of Eq. (49). Explicitly, with $r_x=u$, $r_y=v$ and $q_k=r_{\ell+k}$, take

$$\begin{aligned} q_2&=f_1, & q_8&=f_2, & q_5&=f_3,\\ q_1&=f_1\times u, & q_3&=f_1\times v, & &\\ q_9&=f_2\times u, & q_7&=f_2\times v, & &\\ q_{14}&=v\times q_3, & q_{15}&=v\times q_7, & &\\ q_{19}&=u\times q_1, & q_{23}&=u\times q_9. & & \end{aligned} \tag{52}$$

Faithfulness of a compound hypergraph additionally requires checking that all its rays are distinct and that no orthogonal pair is absent from its two-section graph.

Proof. The overlap bound makes both square roots real. The vectors e_1,e_2,e_3 are mutually orthogonal, so f_1,f_2,f_3 are orthonormal. Lagrange's identity gives

$$(f_i\times u)\cdot(f_i\times v)=u\cdot v-(f_i\cdot u)(f_i\cdot v)=0,$$

because

$$(f_i\cdot u)(f_i\cdot v)=\tfrac{1}{8}(\|e_1\|^2-\cos^2\phi\,\|e_2\|^2)=u\cdot v.$$

Thus $q_1\perp q_3$ and $q_7\perp q_9$; all remaining required orthogonalities follow directly from the cross products in Eq. (52). □

Theorem 13 (Roof-Twin Cascades in $\mathbb{R}^3$)**.** *Starting from the nine-context hypergraph $I_{16,9}$, define*

$$\begin{aligned} G_{27,16}&=I_{16,9}\cup B(18,24;100),\\ G_{41,25}&=G_{27,16}\cup\{\{14,27,26\},\{18,28,26\}\}\\ &\qquad\cup B(27,28;200),\\ G_{41,26}&=G_{41,25}\cup\{\{4,5,105\}\}. \end{aligned}$$

All three admit FORs in $\mathbb{R}^3$ and strong 3-colorings. Their respective values of $(N_c,|\mathcal{S}_{\text{ext}}|,|\mathcal{S}|)$ are $(36,54,82)$, $(216,216,396)$, and $(48,63,107)$. Every two-valued state of $G_{27,16}$ satisfies $s(11)=s(18)$ and $s(10)=s(24)$. In each of the two 41-ray hypergraphs, every two-valued state satisfies

$$\begin{gathered} s(11)=s(18)=s(27),\qquad s(14)=s(28),\\ s(3)=s(26). \end{gathered} \tag{53}$$

In particular,

$$\begin{gathered} U_{\text{TVS}}=\{3,27,28\},\\ s(3)+s(27)+s(28)=1\qquad(\forall s\in\mathcal{S}) \end{gathered} \tag{54}$$

is a pairwise-nonorthogonal two-valued state admissibility context in both $G_{41,25}$ and $G_{41,26}$.

Proof. Binary implication. The original bug $B(10,11;0)$ and the attached $B(18,24;100)$ supply the two exclusivity assumptions for the roof contexts $\{4,11,24\}$ and $\{4,10,18\}$. Lemma 11 gives $s(11)=s(18)$ and $s(10)=s(24)$. Context $E_8=\{3,11,14\}$ therefore yields

$$s(3)+s(14)+s(18)=1, \tag{55}$$

so $s(14)+s(18)\le 1$. The third bug supplies $s(27)+s(28)\le 1$. Applying the same lemma to $\{26,14,27\}$ and $\{26,18,28\}$ gives $s(14)=s(28)$ and $s(18)=s(27)$. Substitution into Eq. (55) proves Eq. (54); comparison with either second-roof context also gives $s(3)=s(26)$. Adding a context only restricts the state space, so these implications hold for both 41-ray cases. This proves universal normalization without enumerating states.

Faithful coordinates. Geometric realizations must be distinguished from the preceding binary argument. A parametrization of $I_{16,9}$ is

$$\begin{aligned} r_1&=(0,1,\alpha), & r_2&=(1,0,0),\\ r_3&=(0,\alpha,-1), & r_5&=(0,0,1),\\ r_7&=(1,0,\beta), & r_8&=(0,1,0),\\ r_9&=(\beta,0,-1), & r_{10}&=(1,-\alpha\beta,\beta),\\ r_{11}&=(\alpha\beta,-1,-\alpha), & r_4&=r_{10}\times r_{11},\\ r_{14}&=r_{11}\times r_3, & r_{15}&=r_7\times r_{11},\\ r_{19}&=r_1\times r_{10}, & r_{23}&=r_{10}\times r_9,\\ r_{18}&=r_4\times r_{10}, & r_{24}&=r_{11}\times r_4. \end{aligned} \tag{56}$$

At $(\alpha,\beta)=(9/32,24/11)$,

$$\begin{aligned} \|r_4\|^2&=(1+\alpha^2)(1+\beta^2)(1+\alpha^2\beta^2),\\ \|r_4\|&=45305/15488. \end{aligned} \tag{57}$$

Put $n=r_4/\|r_4\|$ and $R=nn^{\mathsf{T}}-[n]_\times$, where $[n]_\times w=n\times w$. This is a quarter-turn about n. Both tips lie in $n^\perp$, where $Rw=-n\times w$, so it sends the rays of r_{10},r_{11} to those of r_{18},r_{24}, respectively. Assigning $r_{100+k}=Rr_k$ for the eleven internal bug labels in Eq. (49) gives a rational FOR of $G_{27,16}$ (and integer representatives after rescaling).

At these same rational parameters, take $r_{26}=r_{14}\times r_{18}$, $r_{27}=r_{26}\times r_{14}$, $r_{28}=r_{26}\times r_{18}$ and apply Lemma 12 to the last two rays. Their squared normalized overlap is

$$\frac{21159411204096}{3980389958002705}<\frac{1}{9}.$$

This gives a FOR of $G_{41,25}$. The rotation and cross-product formulas establish all intended orthogonalities algebraically; outward-rounded interval evaluation at 70 decimal digits certifies every other inner product to be nonzero and every pair of rays to be distinct. The smallest certified absolute normalized nonedge inner product exceeds 0.00155.

For $G_{41,26}$, a simpler, separate realization uses $\alpha = \beta = 1$. Choose the ray representatives $r_4 = (1,1,0)$, $r_{14} = (2,1,1)$, $r_{18} = (1,-1,-2)$, $r_{24} = (1,-1,2)$. For the second bug use Eq. (52) with $f_1 = (1,1,\sqrt{2})$, $f_2 = (1,1,-\sqrt{2})$ (normalization is immaterial for rays). For the second roof use

$$\begin{aligned} r_{26} &= (-1,5,-3), \\ r_{27} &= (8,-5,-11), \\ r_{28} &= (-13,-5,-4). \end{aligned} \tag{58}$$

Their normalized overlap has magnitude $1/6$. For the third bug, Eq. (52) applies with

$$\begin{aligned} f_{1,2} = \sqrt{35}\,(3,0,-1) \\ \pm \big(\sqrt{15}\,(1,2,3) + 2(-1,5,-3)\big). \end{aligned} \tag{59}$$

Exact arithmetic in $\mathbb{Q}(\sqrt{2},\sqrt{15},\sqrt{35})$ verifies all 820 ray pairs: rays are distinct, and orthogonality is precisely adjacency in $G_{41,26}$. In particular,

$$r_3 \cdot r_{27} = 6, \quad r_3 \cdot r_{28} = -1, \quad r_{27} \cdot r_{28} = -35,$$

so the target triple is pairwise nonorthogonal. At these symmetric parameters, deleting $\{4,5,105\}$ leaves three unrecorded orthogonal pairs, namely $(4,5),(4,105),(5,105)$. Thus these symmetric coordinates are faithful for $G_{41,26}$, whereas $G_{27,16}$ and $G_{41,25}$ use the separate rational-parameter construction above.

Finite counts. Exhaustive enumeration, independently checked by a second implementation, gives the stated state and normalized-coloring counts. The coordinate and state certificate supplies the full contexts, ray coordinates, assignments, and verification routines. Enumeration certifies the counts and geometry; the two applications of Lemma 11 establish normalization on *all* two-valued states. □

The separation from affine and operator normalization can be seen directly. In the symmetric FOR of $G_{41,26}$, prepare $\psi = \hat{r}_3 = (0,1,-1)/\sqrt{2}$. Since $\|r_{27}\|^2 = \|r_{28}\|^2 = 210$, its Born probabilities give

$$\begin{aligned} p_\psi(3) + p_\psi(27) + p_\psi(28) &= 1 + \frac{36}{420} + \frac{1}{420} \\ &= \frac{457}{420} \neq 1. \end{aligned} \tag{60}$$

This weight lies in $\mathcal{W}(G_{41,26}) \subset \mathcal{A}(G_{41,26})$, while every convex combination of two-valued states has target sum one. Hence $\operatorname{aff}\mathcal{S}(G_{41,26}) \subsetneq \mathcal{A}(G_{41,26})$ and $P_3 + P_{27} + P_{28} \neq I_3$, in agreement with Proposition 7. The uniform weight $p(v) = 1/3$ satisfies the target normalization and every twin identity; it is not a separating witness. Indeed, any strong 3-coloring expresses it as the average of its three color-class states.

Proposition 14 (Binary Restrictions in the Yu–Oh Examples)**.** *In the fixed Yu–Oh representation, the unique triple of pairwise nonorthogonal rays satisfying* $\sum_{v\in U} s(v) \le 1$ *on every state of* $H_{21,13}$ *is* $U = \{10,11,13\}$. *No proper context subhypergraph has such a triple. Moreover, impose* $s(a) + s(b) \le 1$ *for the following pairs* (a,b)*:*

$$\begin{aligned} H_{20,12}: &\quad (18,24),(20,21), \\ H_{21,13}: &\quad (18,24),(15,21),(19,20). \end{aligned} \tag{61}$$

These filters retain exactly the original extendable states: 10 *of* 25 *states for* $H_{20,12}$*, and* 6 *of* 21 *for* $H_{21,13}$*. Thus the two, respectively* 21*, independent chromatic admissible triples are normalized on these restricted state families.*

Proof. Enumeration of the 21 states and the independent triples of $H_{21,13}$ gives the unique exclusive triple $\{10,11,13\}$. For each of the 13 single-context deletions, the certificate supplies a state with hub sum greater than one. Every proper context subhypergraph is contained in one of these deletions, and removing further context equations preserves a violating assignment. Any triple of pairwise nonorthogonal rays exclusive on a hypergraph with fewer contexts would also be exclusive on the full hypergraph, hence would have to be the same hub triple. This proves the minimum by 13 deletion witnesses and monotonicity. Direct filtering of the 25 and 21 original states gives exactly the 10 and 6 extendable states specified above. □

These filters are statements about binary state families. Abstract Specker-bug attachments can enforce the listed inequalities, because each allowed tip assignment $(0,0),(0,1),(1,0)$ extends to a bug state. A faithful three-dimensional realization of all simultaneous attachments would require a separate check of ray injectivity and unintended orthogonalities. No such geometric extension is asserted here.

VI. FOUR-DIMENSIONAL CHROMATIC RELATIONS IN $\mathbb{R}^4$

In dimension $d = 4$, related chromatic relations occur both in common-apex lifts and in strictly linear 4-uniform hypergraphs. The lifts $B^{(4)}_{14,7}$ and $D^{(4)}_{17,9}$ prepend a singleton apex column to every Context–Color Matrix [9, Proposition 2]; they preserve the three-dimensional universally rainbow pairs and color-multiset-equivalent pairs. The strictly linear examples ($|e \cap e'| \le 1$, girth $g \ge 3$) are $L^{(4)}_{18,7}$ ($|V| = 18$, $|E| = 7$, $N_c = 144$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 68/68$) and $L^{(4)}_{22,9}$ ($|V| = 22$, $|E| = 9$,

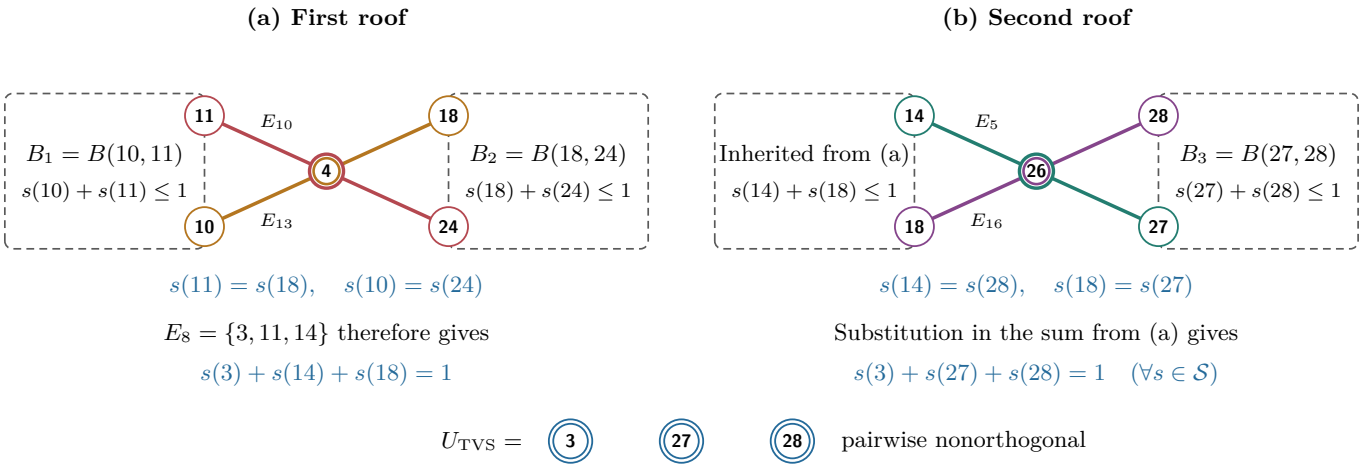


FIG. 6. The two applications of the roof-twin lemma in Theorem 13. Each solid trace contains one complete three-vertex roof context; their shared numbered circle is the roof vertex. Dashed boxes compress seven-context bugs, except the left box in (b), which records the inequality inherited from (a). Box boundaries are not contexts, and internal bug vertices are omitted. In (a), $\{4, 11, 24\}$ and $\{4, 10, 18\}$ force the first twins; context $E_8 = \{3, 11, 14\}$ then gives the sum shown below. In (b), $\{26, 14, 27\}$ and $\{26, 18, 28\}$ force the second twins, yielding normalization on U_{TVS}. The three separate double-outlined markers identify that target triple without adding a context or orthogonality relation. Repeated numbers in the panels refer to the same rays. This is a logical schematic common to $G_{41,25}$ and $G_{41,26}$; the extra context $\{4, 5, 105\}$ of the latter is omitted. The proof and coordinate certificate establish the full faithful geometry independently of this drawing.

$N_c = 216$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}| = 92/158$) [8, 9]. The complete rainbow quadruples exhibited below are partially nonorthogonal, rather than independent admissibility contexts.

In $L^{(4)}_{22,9}$ (Fig. 7), the 9 four-element contexts are

$$\begin{aligned} C_1 &= \{1, 2, 4, 3\}, & C_2 &= \{5, 8, 7, 6\}, \\ C_3 &= \{4, 9, 10, 8\}, & C_4 &= \{11, 1, 12, 5\}, \\ C_5 &= \{11, 14, 13, 2\}, & C_6 &= \{15, 3, 16, 7\}, \\ C_7 &= \{15, 17, 18, 6\}, & C_8 &= \{11, 21, 22, 19\}, \\ C_9 &= \{13, 20, 15, 19\}, \end{aligned} \tag{62}$$

with exact integer FOR in $\mathbb{R}^4$ given by

$$\begin{aligned} r_1 &= (1, 0, 0, 0), & r_2 &= (0, 1, 0, 0), \\ r_3 &= (0, 0, 1, 0), & r_4 &= (0, 0, 0, 1), \\ r_5 &= (0, 1, -1, -1), & r_6 &= (4, -5, -11, 6), \\ r_7 &= (1, -4, 0, -4), & r_8 &= (4, 1, 1, 0), \\ r_9 &= (1, -5, 1, 0), & r_{10} &= (2, -1, -7, 0), \\ r_{11} &= (0, 0, 1, -1), & r_{12} &= (0, 2, 1, 1), \\ r_{13} &= (1, 0, 4, 4), & r_{14} &= (8, 0, -1, -1), \\ r_{15} &= (4, 2, 0, -1), & r_{16} &= (4, -5, 0, 6), \\ r_{17} &= (5, -9, 7, 2), & r_{18} &= (2, 113, 77, 234), \\ r_{19} &= (16, -33, -2, -2), & r_{20} &= (4, 2, -21, 20), \\ r_{21} &= (22, 12, -11, -11), & r_{22} &= (129, 44, 153, 153). \end{aligned} \tag{63}$$

Across all $N_c = 216$ normalized 9×4 Context–Color Matrices of $L^{(4)}_{22,9}$, the base subhypergraph $L^{(4)}_{18,7}$ $(C_1, \ldots, C_7)$ forces the universally rainbow pair $c(11) \neq c(15)$. In $C_5 = \{11, 14, 13, 2\}$ and $C_9 = \{13, 20, 15, 19\}$, this propagates to 7 further nonorthogonal rainbow pairs

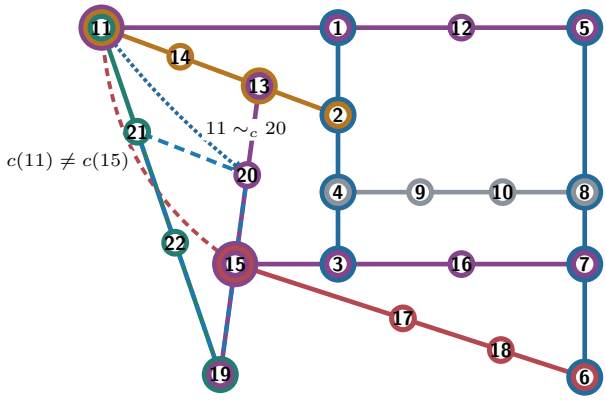


FIG. 7. The strictly linear 22-ray, 9-context hypergraph $L^{(4)}_{22,9}$ in $\mathbb{R}^4$ [Eqs. (62)–(63)], with all 9 four-element contexts drawn as straight lines. Shown are the universally rainbow pair $c(11) \neq c(15)$, the monochromatic pair $11 \sim_c 20$, and the complete rainbow quadruple $\{19, 20, 21, 22\}$ (blue dashed polygon).

$$\begin{aligned} &\{11, 15\},\ \{1, 20\},\ \{2, 20\},\ \{5, 20\}, \\ &\{12, 20\},\ \{14, 20\},\ \{15, 21\},\ \{15, 22\}, \end{aligned} \tag{64}$$

the monochromatic nonorthogonal pair $11 \sim_c 20$ ($c(11) = c(20)$), and 4 partially nonorthogonal complete rainbow quadruples ($\{c(u_1), c(u_2), c(u_3), c(u_4)\} = \{0, 1, 2, 3\}$ in all 216 normalized colorings):

$$\begin{aligned} &\{19, 20, 21, 22\}, \quad \{1, 5, 12, 20\}, \\ &\{2, 13, 14, 20\}, \quad \{11, 13, 15, 19\}. \end{aligned} \tag{65}$$

TABLE II. Census of chromatic and two-valued state admissibility contexts and related relations across 19 quantum hypergraphs in $\mathbb{R}^3$, $\mathbb{C}^3$, and $\mathbb{R}^4$. Listed are the Hilbert space $\mathbb{F}^d$, vertex/context counts, normalized coloring count N_c, extendable-to-total state ratio, universally rainbow nonorthogonal pairs, chromatic-twin pairs, chromatic admissible contexts on $\mathcal{S}_{\text{ext}}(H)$ (with additional partially nonorthogonal complete rainbow d-tuples in parentheses), two-valued state admissibility contexts on $\mathcal{S}(H)$ (with additional partially nonorthogonal two-valued complete d-tuples in parentheses), and non-twin color-multiset-equivalent pairs of sizes $k = 2, 3$. The paired columns omit pairs of two already complete admissible contexts; both independent triples of $H_{53,34}$ are normalized on all two-valued states.

Config.	$\mathbb{F}^d$	$(\lvert V\rvert, \lvert E\rvert)$	N_c	$\lvert\mathcal{S}_{\text{ext}}\rvert/\lvert\mathcal{S}\rvert$	Rainbow pairs	Twin pairs	Chromatic admissibility	Two-valued admissibility	Paired $k=2$	Paired $k=3$
$M_{15,8}$	$\mathbb{R}^3$	$(15, 8)$	16	24/24	0	0	0 (0)	0 (0)	0	4
H_{20}	$\mathbb{C}^3$	$(20, 12)$	12	26/26	0	0	0 (0)	0 (0)	1	0
$B_{13,7}$	$\mathbb{R}^3$	$(13, 7)$	6	12/14	1	0	0 (0)	0 (0)	0	1
$D_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	7	14/21	1	0	0 (0)	0 (0)	0	0
$I_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	6	12/22	10	2	0 (6)	0 (0)	2	3
$H_{17,10}$	$\mathbb{R}^3$	$(17, 10)$	4	10/18	16	2	0 (6)	0 (0)	2	3
$T_{19,11}$	$\mathbb{R}^3$	$(19, 11)$	7	14/25	8	2	0 (4)	0 (0)	1	2
$H_{20,12}$	$\mathbb{R}^3$	$(20, 12)$	4	10/25	28	4	**2** (12)	0 (0)	8	24
$H_{21,13}$	$\mathbb{R}^3$	$(21, 13)$	2	6/21	66	21	**21** (63)	0 (0)	36	0
$G_{27,16}$	$\mathbb{R}^3$	$(27, 16)$	36	54/82	18	2	0 (10)	0 (10)	4	10
$G_{41,25}$	$\mathbb{R}^3$	$(41, 25)$	216	216/396	46	6	**1** (29)	**1** (29)	10	29
$G_{41,26}$	$\mathbb{R}^3$	$(41, 26)$	48	63/107	60	6	**1** (29)	**1** (29)	10	308
$H_{53,34}$	$\mathbb{R}^3$	$(53, 34)$	128	128/224	66	4	**2** (23)	**2** (23)	12	416
$R_{21,12}$	$\mathbb{C}^3$	$(21, 12)$	32	48/54	0	0	0 (0)	0 (0)	0	12
$R_{25,15}$	$\mathbb{C}^3$	$(25, 15)$	24	48/81	0	0	0 (0)	0 (0)	0	12
$B^{(4)}_{14,7}$	$\mathbb{R}^4$	$(14, 7)$	6	13/15	1	0	0 (0)	0 (0)	0	1
$D^{(4)}_{17,9}$	$\mathbb{R}^4$	$(17, 9)$	7	15/22	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{18,7}$	$\mathbb{R}^4$	$(18, 7)$	144	68/68	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{22,9}$	$\mathbb{R}^4$	$(22, 9)$	216	92/158	8	1	0 (4)	0 (0)	0	0

VII. MASTER CENSUS AND CONCLUSION

Table II summarizes the invariants and admissibility relations of all 19 configurations, including $H_{53,34}$. Its two independent chromatic admissible triples, $\{a, b, u\}$ and $\{a', b', u\}$, also satisfy the stronger two-valued state admissibility condition. The paired columns omit equivalences between two already complete admissibility contexts, as specified in Definition 2.

The central phenomenon is enforced exclusivity and completeness on a single independent set, stratified into a three-tier hierarchy by the underlying state family:

1. *Chromatic admissible contexts on $\mathcal{S}_{\text{ext}}(H)$:* Rainbow in every strong d-coloring, or equivalently normalized ($\sum_{u\in U} s(u) = 1$) on every extendable color-class state. In the Yu–Oh family, pairwise-nonorthogonal chromatic admissibility triples first appear at 12 contexts ($H_{20,12}$, 2 triples) and reach 21 triples in $H_{21,13}$ (Theorem 10), including the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ and the 60° basis $U_{\text{core}} = \{1, 4, 7\}$.

2. *Two-valued admissibility contexts on $\mathcal{S}(H)$:* Normalized ($\sum_{u\in U} s(u) = 1$) on *every* admissible two-valued state $s \in \mathcal{S}(H)$. Even though no linear combination of context equations can force a nonorthogonal d-tuple to sum to 1 on $\mathcal{A}(H)$ (Proposition 7), $\{0, 1\}$-integrality on Specker bugs across two-context roofs enforces biconditional two-valued twins $s(x) = s(x')$ (Lemma 11), causing $\operatorname{aff}\mathcal{S}(H) \subsetneq \mathcal{A}(H)$ and yielding the pairwise-nonorthogonal two-valued state admissibility context $U_{\text{TVS}} = \{3, 27, 28\}$ in $G_{41,25}$ and $G_{41,26}$ (Theorem 13). Proposition 14 gives additional binary state restrictions for $H_{20,12}$ and $H_{21,13}$; faithful geometric realizations of simultaneous gadget attachments are not claimed. The 53-ray construction of Section IV gives two further independent two-valued state admissibility contexts and an explicit Born-state total 8/9, with a full hypergraph drawing and exact coordinate certificate.

3. *Operator and affine rigidity on $\mathcal{W}(H)$ and $\mathcal{A}(H)$:* No nonorthogonal d-tuple can resolve the identity $\sum_{u\in U} P_u = I_d$ or be normalized on $\mathcal{A}(H)$. Meanwhile, balanced context certificates yield paired affine pseudocontexts ($M_{15,8}$ and H_{20}), which imply color-multiset equivalence on $\mathcal{S}_{\text{ext}}(H)$ without requiring single-set completeness.

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